

Data-Driven Investment Intelligence Using NIFTY-50 Market Data

Technical Report

Dataset coverage: 2000-01-03 to 2021-04-30 | 50 companies | 235,191 daily records

This report documents the design, methodology, and results of an AI-powered investment intelligence platform built on the NIFTY-50 historical market dataset (Jan 2000–Apr 2021). The platform delivers a Stock Predictor Engine, a Portfolio Construction Module for Conservative, Balanced and Aggressive investor profiles, a Risk Assessment Module, and additional Market Anomaly Detection capabilities, all delivered via an interactive Streamlit dashboard.

1. Exploratory Data Analysis (EDA)

The provided dataset (**NIFTY50_all.csv**) contains **235,191** daily OHLCV records for **50** companies spanning **2000-01-03** to **2021-04-30**. Each record includes Open, High, Low, Close, Volume, Turnover, Trades, Deliverable Volume and %Deliverable.

1.1 Symbol Consolidation

Several constituents changed their NSE trading symbol over the dataset's history due to corporate renames, mergers, or demergers (e.g. **INFOSYSTCH**→**INFY**, **HEROHONDA**→**HEROMOTOCO**, **MUNDRAPORT**→**ADANI PORTS**, **TELCO**→**TATAMOTORS**, **UTIBANK**→**AXISBANK**, and others). These were consolidated into a single continuous price series per company, yielding exactly 50 unique, continuous symbols.

1.2 Sector Mapping

The dataset does not include sector/industry metadata. A manual sector and industry classification was curated for all 50 constituents (Banking, IT Services, Pharmaceuticals, Energy, FMCG, Automobile, Manufacturing, Telecom, etc.) based on publicly known NSE classifications, enabling sector-level aggregation throughout the platform.

1.3 Data Quality

Rows with non-positive prices were removed; duplicate (Symbol, Date) records were dropped; missing Trades / Deliverable Volume / %Deliverable values (present mainly in early years before these fields were reported) were filled with 0. Daily simple and log returns were derived per symbol.

1.4 Market-Wide Trends

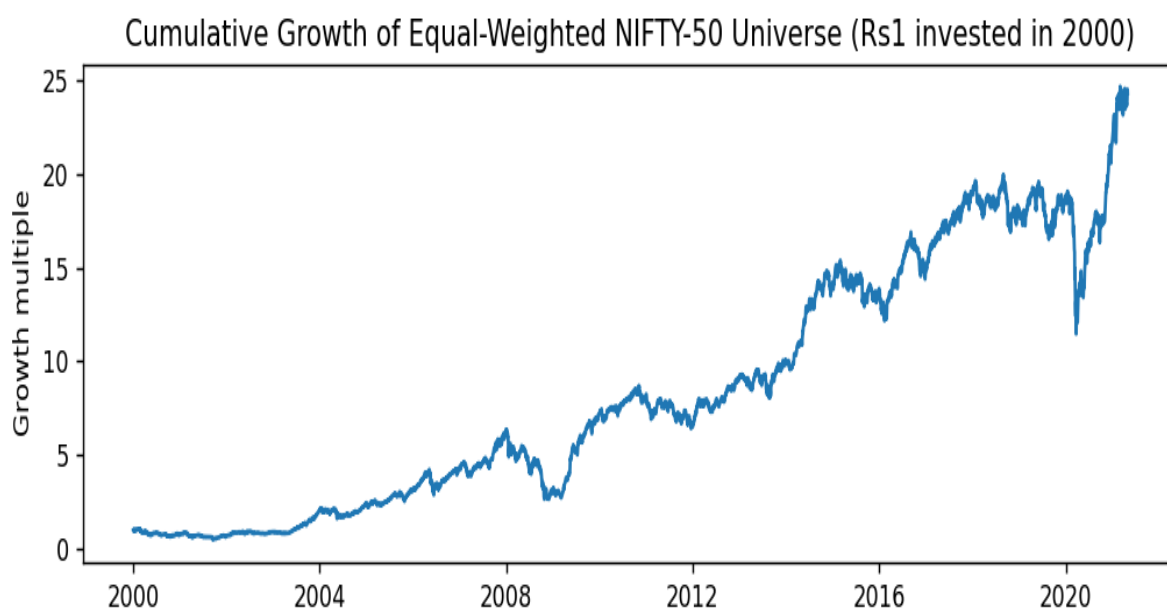


Figure 1.1 — Cumulative growth of an equal-weighted basket of all 50 NIFTY constituents, illustrating long-run market cycles (2008 crash, 2020 COVID crash and recovery, etc.).

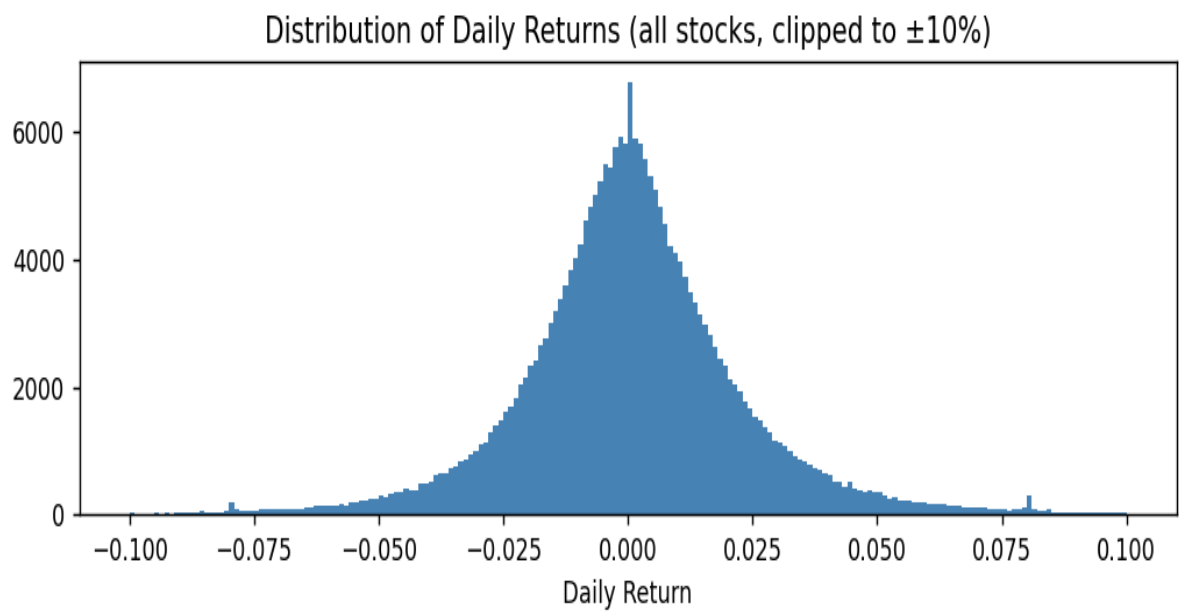


Figure 1.2 — Distribution of daily returns across all stocks, showing the typical fat-tailed, leptokurtic shape of equity returns.

1.5 Risk-Return Landscape

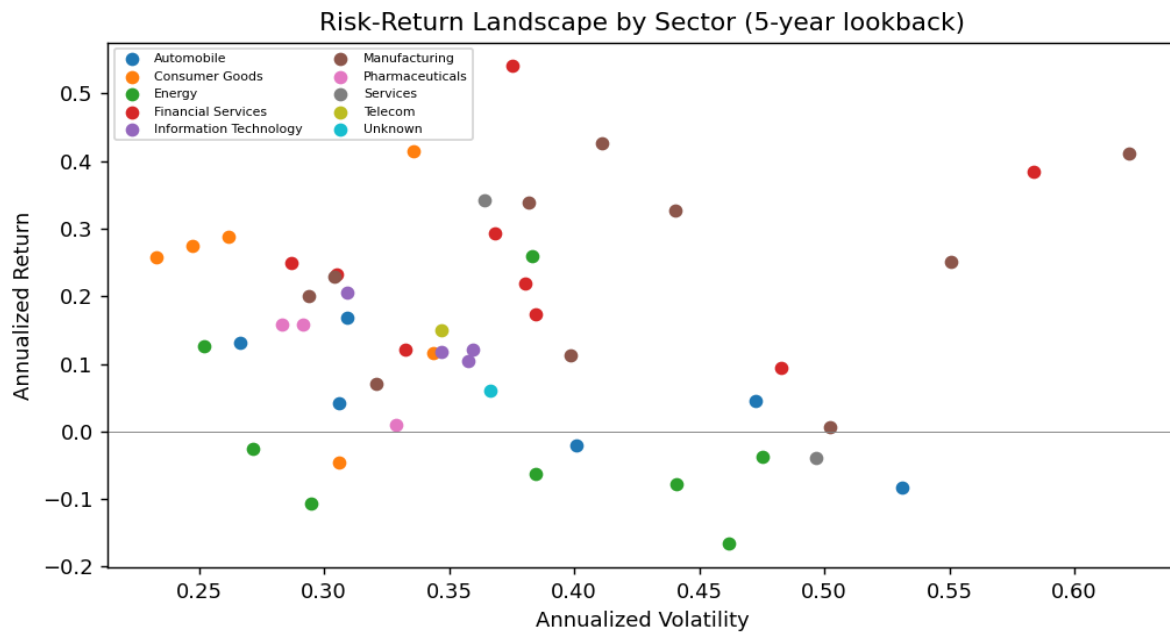


Figure 1.3 — 5-year annualized return vs. volatility for each constituent, colored by sector.

1.6 Top / Bottom Performers (5-Year Lookback)

Symbol	Annualized Return	Annualized Volatility	Sharpe Ratio
BAJAJFINSV	54.04%	37.53%	99.23%
HINDALCO	42.62%	41.12%	71.79%
TITAN	41.50%	33.59%	85.56%
SESAGOA	41.17%	62.17%	45.85%
BAJFINANCE	38.36%	58.37%	45.39%

Top 5 performers (table above) were dominated by financial services and consumer discretionary names (Bajaj Finserv, Titan), while the **bottom 5** (table below) were concentrated in energy / commodity names (IOC, Coal India, ONGC, GAIL), reflecting the structural de-rating of PSU energy stocks over the period.

Symbol	Annualized Return	Annualized Volatility	Sharpe Ratio
IOC	-16.60%	46.20%	-52.26%
COALINDIA	-10.73%	29.47%	-58.85%
EICHERMOT	-8.28%	53.11%	-27.57%
GAIL	-7.80%	44.09%	-32.04%
ONGC	-6.25%	38.44%	-32.41%

1.7 Sector Performance

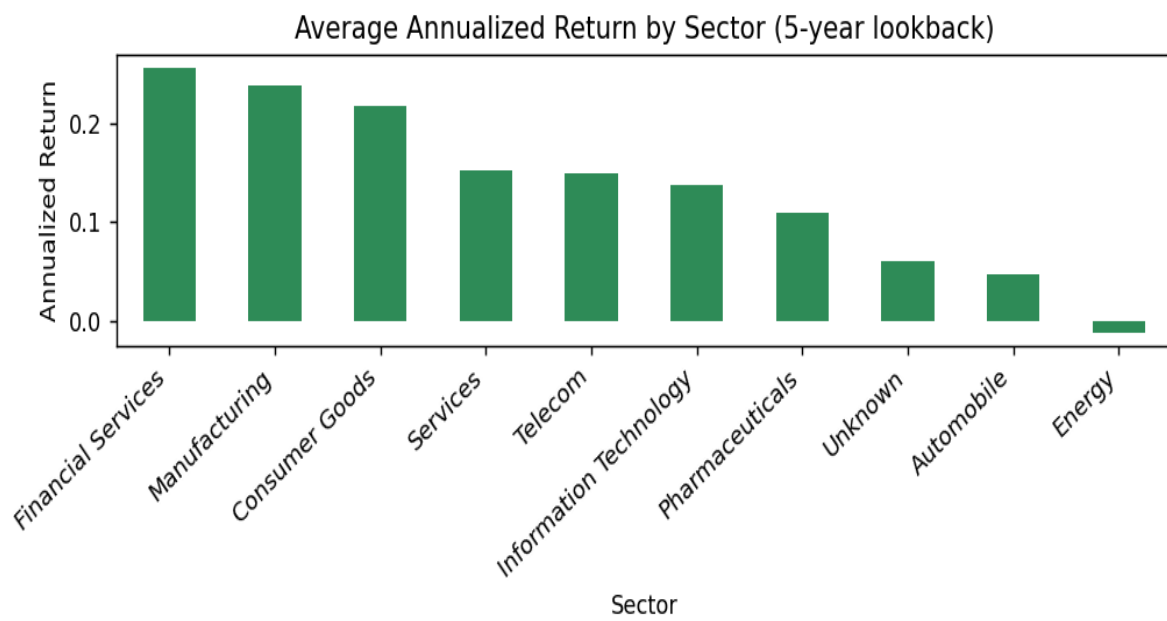


Figure 1.4 — Average 5-year annualized return by sector.

2. Feature Engineering

For each stock, a comprehensive technical-indicator feature set is computed (see `src/indicators.py`), used both for the predictor engine and for dashboard visualizations:

- **Trend:** Simple Moving Averages (5/10/20/50/100/200-day) and Exponential Moving Averages (12/26/50-day).
- **Momentum:** Relative Strength Index (RSI-14); Rate-of-change momentum over 5/10/20/60 days.
- **Trend/Momentum combo:** MACD line, signal line and histogram (12/26/9).
- **Volatility:** Bollinger Bands (20-day, $\pm 2\sigma$), Bollinger %B and band width; rolling annualized volatility of log returns (10/20/30-day windows).
- **Volume:** rolling average volume and volume ratio (current vs. 10/20-day average) to capture unusual trading activity.
- **Returns:** daily simple and log returns.

All indicators are computed per-symbol on chronologically sorted data, with rolling windows producing NaNs for the initial warm-up period (dropped before model training).

2.1 Example: RELIANCE Indicator Suite

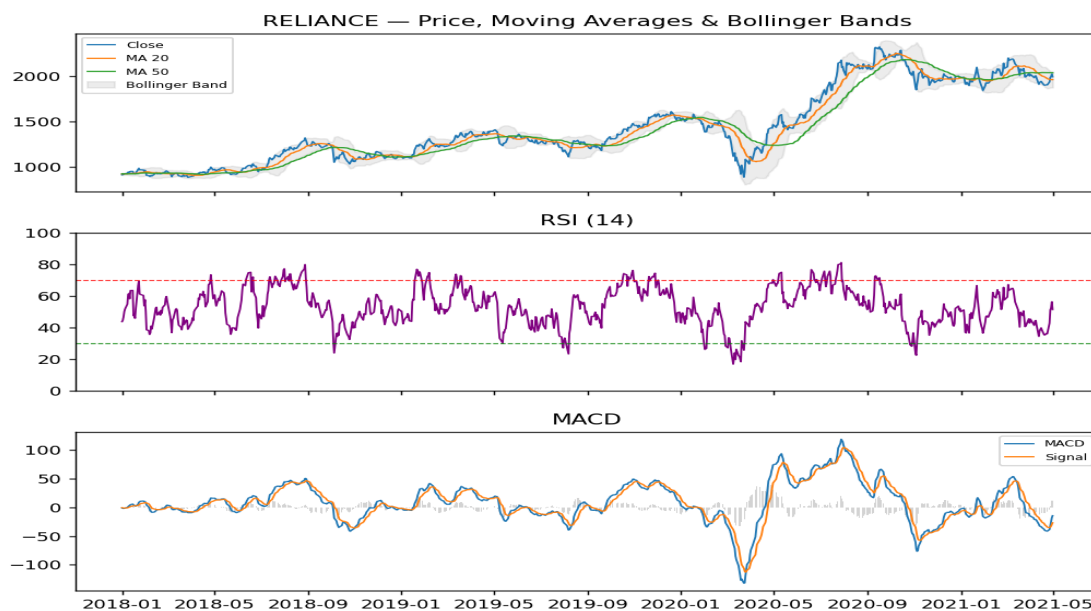


Figure 2.1 — Price with moving averages and Bollinger Bands (top), RSI-14 (middle), and MACD (bottom) for RELIANCE, 2018–2021.

3. Methodology

The platform follows a modular pipeline implemented in Python (pandas, NumPy, scikit-learn, SciPy) and exposed through a Streamlit dashboard:

- **Data Layer** (*data_utils.py*): loads the raw CSV, consolidates renamed symbols, applies sector mapping, cleans data, and computes daily returns.
- **Feature Layer** (*indicators.py*): computes the full technical-indicator suite per symbol.
- **Predictor Layer** (*predictor.py*): builds supervised learning datasets and trains regression and classification models.
- **Risk Layer** (*risk.py*): computes risk and risk-adjusted performance metrics for individual stocks and portfolios.
- **Portfolio Layer** (*portfolio.py*): performs mean-variance optimization over risk-filtered candidate universes for three investor profiles.
- **Anomaly Layer** (*anomaly.py*): flags statistically unusual return/volume days and major drawdown episodes.
- **Presentation Layer** (*app.py*): a six-page Streamlit dashboard tying all modules together with interactive Plotly visualizations.

3.1 Train/Test Strategy

All predictive models use a **chronological (walk-forward) 80/20 split** — the most recent 20% of each stock's history is held out as a test set. This avoids look-ahead bias inherent to random shuffling of time-series data and gives a realistic estimate of out-of-sample performance.

4. Model Architecture — Stock Predictor Engine

For a chosen forecast horizon h (default 5 trading days), two targets are derived for each day t : the **future return** $\text{FutureReturn}_t = (\text{Close}_{t+h} - \text{Close}_t) / \text{Close}_t$, and the **direction** $\text{Direction}_t = 1$ if $\text{FutureReturn}_t > 0$ else 0.

- **Regression model**: Random Forest Regressor (300 trees, max depth 8, min_samples_leaf=5) predicts FutureReturn from the 24 technical-indicator features listed in Section 2.
- **Classification model**: Random Forest Classifier (same hyperparameters) predicts Direction (up/down).
- An alternative **Linear Regression** mode is available for the regression task (with the RF classifier retained for direction), offering a simpler, more interpretable baseline.

4.1 Evaluation Metrics

Each model is evaluated on its held-out chronological test set using:

- **MAE** — Mean Absolute Error of predicted vs. actual future return.
- **RMSE** — Root Mean Squared Error of predicted vs. actual future return.
- **R²** — coefficient of determination for the regression model.
- **Directional Accuracy** — % of test days where the predicted sign of return (from either the regressor or the dedicated classifier) matches the actual sign.

4.2 Results (5-day horizon, sample of 8 large-cap stocks)

Symbol	MAE	RMSE	R2	Dir_Acc_Reg	Dir_Acc_Clf	n_test
RELIANCE	0.037	0.060	-0.072	0.557	0.536	1049.000
TCS	0.035	0.057	-0.083	0.493	0.504	815.000
HDFCBANK	0.030	0.054	-0.292	0.544	0.540	1049.000
INFY	0.031	0.053	-0.043	0.523	0.515	1049.000
ITC	0.028	0.039	-0.050	0.473	0.460	1049.000
MARUTI	0.035	0.048	-0.047	0.480	0.510	873.000
SBIN	0.040	0.057	-0.034	0.494	0.465	1049.000
SUNPHARMA	0.036	0.050	-0.045	0.502	0.518	1049.000

On average across this sample, $MAE \approx 0.034$, $RMSE \approx 0.052$, $R^2 \approx -0.083$, and directional accuracy $\approx 50.6\%$ (classifier) / 50.8% (regression sign). The negative R^2 values indicate the regression model does not explain variance in 5-day forward returns better than the historical mean — consistent with the well-documented difficulty of short-horizon equity return prediction from price/volume data alone (efficient-market behavior). Directional accuracy hovering just above 50% suggests a small but consistent edge over a coin-flip baseline for some stocks (e.g. HDFCBANK at 54%), but not a reliable, exploitable signal on its own.

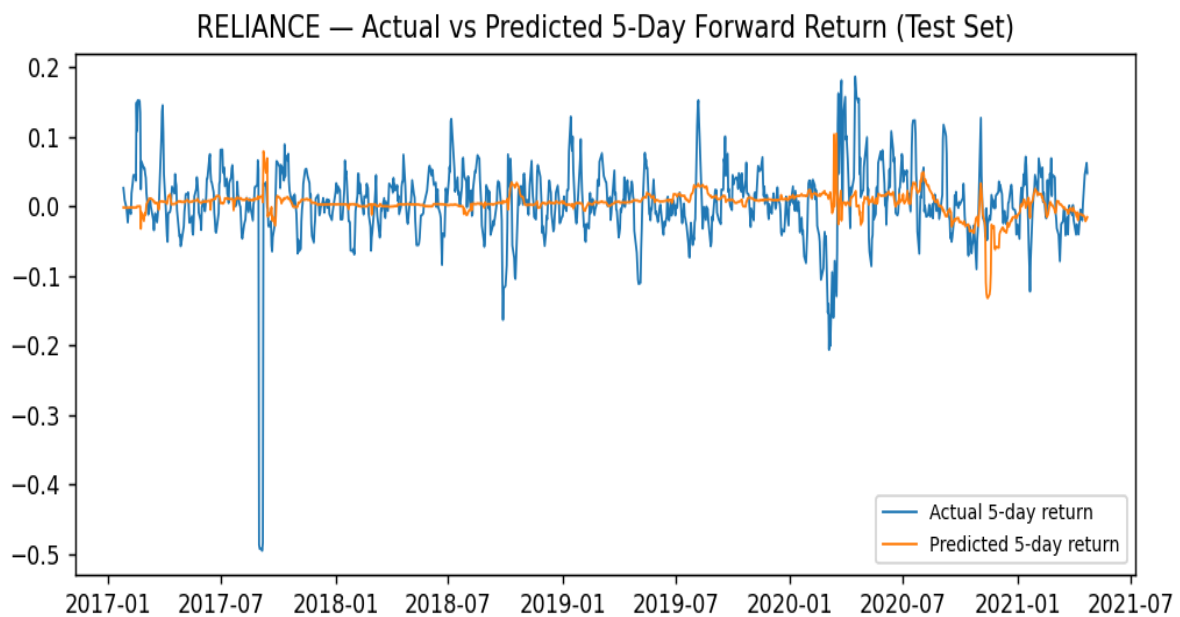


Figure 4.1 — RELIANCE: actual vs. predicted 5-day forward return on the held-out test period. Predictions are heavily shrunk toward zero relative to the noisy actuals, a typical Random-Forest behavior on low-signal targets.

4.3 Latest Forecast Example (RELIANCE)

As of 2021-04-23, with last close ■1904.35, the model projects a 5-day return of -1.60% (target price ■1873.79), while the direction classifier separately estimates a 52% probability of an upward move. Note that the two models can occasionally disagree in sign (as here) — this is reported transparently in the dashboard rather than artificially reconciled, since it reflects genuine model uncertainty near the 50% decision boundary.

5. Risk Assessment Methodology

For each stock (and for constructed portfolios), the following metrics are computed from daily returns r_t and the price series P_t , annualized using 252 trading days:

- **Annualized Return** = $(1 + \text{mean}(r))^{252} - 1$
- **Annualized Volatility** = $\text{std}(r) \times \sqrt{252}$
- **Sharpe Ratio** = $\text{mean}(r - r_f/252) / \text{std}(r - r_f/252) \times \sqrt{252}$, with $r_f = 6\%$ (adjustable in the dashboard)
- **Sortino Ratio** = same as Sharpe but normalizing by the standard deviation of *negative* excess returns only (downside risk)
- **Maximum Drawdown** = worst peak-to-trough decline in cumulative price
- **Calmar Ratio** = Annualized Return / |Maximum Drawdown|
- **Value at Risk (95% / 99%)** = empirical (historical) percentile of the daily return distribution
- **Beta** = $\text{covariance}(\text{stock returns}, \text{equal-weighted NIFTY-50 returns}) / \text{variance}(\text{market returns})$

A simple **risk classification** buckets each stock by annualized volatility: <25% → **Low**, 25–40% → **Medium**, >40% → **High**. This classification drives the candidate universe for portfolio construction (Section 6).

5.1 Example: Risk Profile of Selected Large-Caps

Symbol	Annualized Return	Annualized Volatility	Sharpe Ratio	Sortino Ratio	Max Drawdown	Risk Level
RELIANCE	0.195	0.382	0.310	0.358	-0.790	Medium
TCS	0.164	0.377	0.244	0.252	-0.821	Medium
HDFCBANK	0.220	0.381	0.365	0.388	-0.842	Medium
INFY	0.030	0.477	-0.063	-0.061	-0.969	High
ITC	0.085	0.391	0.054	0.053	-0.941	Medium
MARUTI	0.308	0.345	0.605	0.898	-0.613	Medium
SBIN	0.167	0.426	0.222	0.256	-0.957	High
SUNPHARMA	0.106	0.462	0.088	0.086	-0.907	High

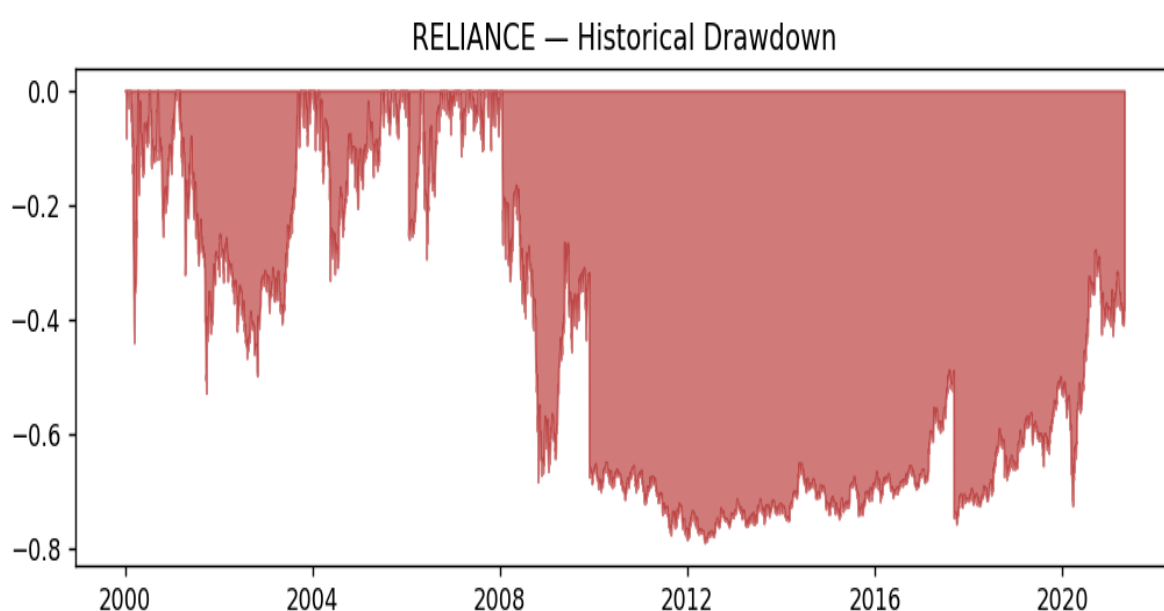


Figure 5.1 — RELIANCE historical drawdown from running peak. Maximum drawdown of -79% occurred during the 2008 global financial crisis.

6. Portfolio Construction Logic

Portfolios are built using **Modern Portfolio Theory (mean-variance optimization)**, implemented with SciPy's SLSQP solver (*portfolio.py*), in three steps:

- **Step 1 — Risk Filtering:** each stock's 5-year annualized return, volatility, Sharpe ratio and risk classification are computed (Section 5). The candidate universe is restricted to stocks whose risk level matches the investor profile.
- **Step 2 — Candidate Ranking:** candidates are ranked by Sharpe ratio (Conservative, Balanced) or by historical return (Aggressive), and the top 15 are retained.
- **Step 3 — Optimization:** weights are solved subject to $\sum w_i = 1$, $0 \leq w_i \leq w_{\max}$, to:
 - **Conservative:** *minimize portfolio variance* — candidate universe restricted to **Low-risk** stocks only; $w_{\max} = 25\%$.
 - **Balanced:** *maximize the Sharpe ratio* — candidate universe is **Low + Medium-risk** stocks; $w_{\max} = 20\%$.
 - **Aggressive:** *maximize expected return* — candidate universe spans **Low + Medium + High-risk** stocks; $w_{\max} = 30\%$.

6.1 Resulting Allocations

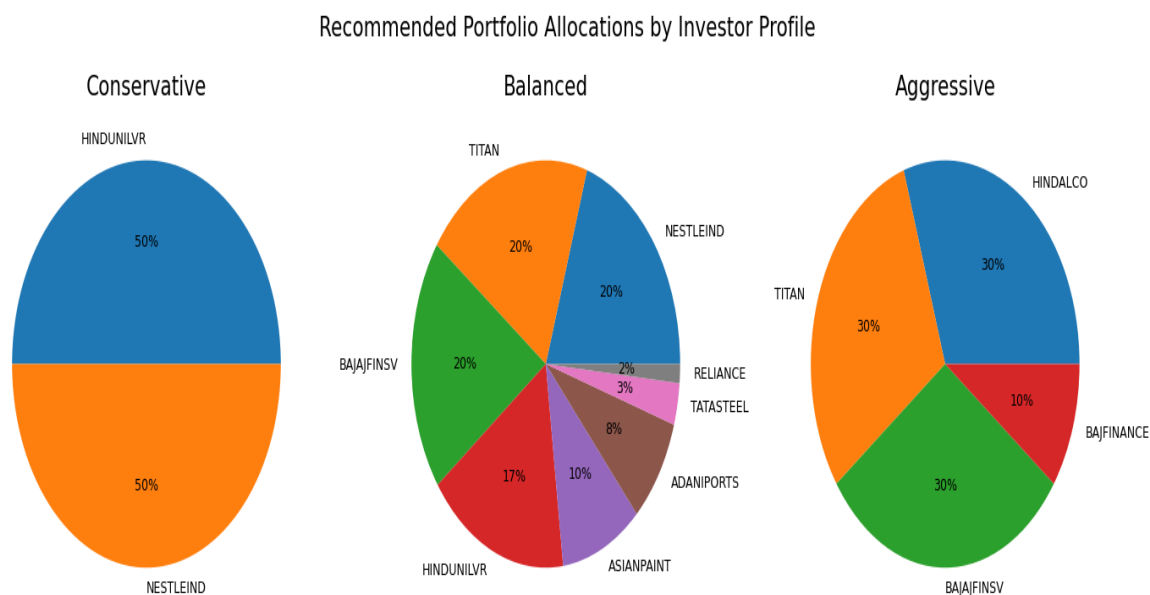


Figure 6.1 — Recommended allocations for the three investor profiles (5-year lookback).

Conservative: Expected annual return 26.7%, expected annual volatility 20.7%, Sharpe ratio 0.85.

Balanced: Expected annual return 35.8%, expected annual volatility 20.4%, Sharpe ratio 1.21.

Aggressive: Expected annual return 45.2%, expected annual volatility 28.8%, Sharpe ratio 1.09.

7. Explainability Techniques

7.1 Feature Importance (Predictor Engine)

Random Forest **impurity-based feature importances** are extracted from the trained regression model and surfaced in the dashboard, showing which technical indicators most influence each stock's return forecast.

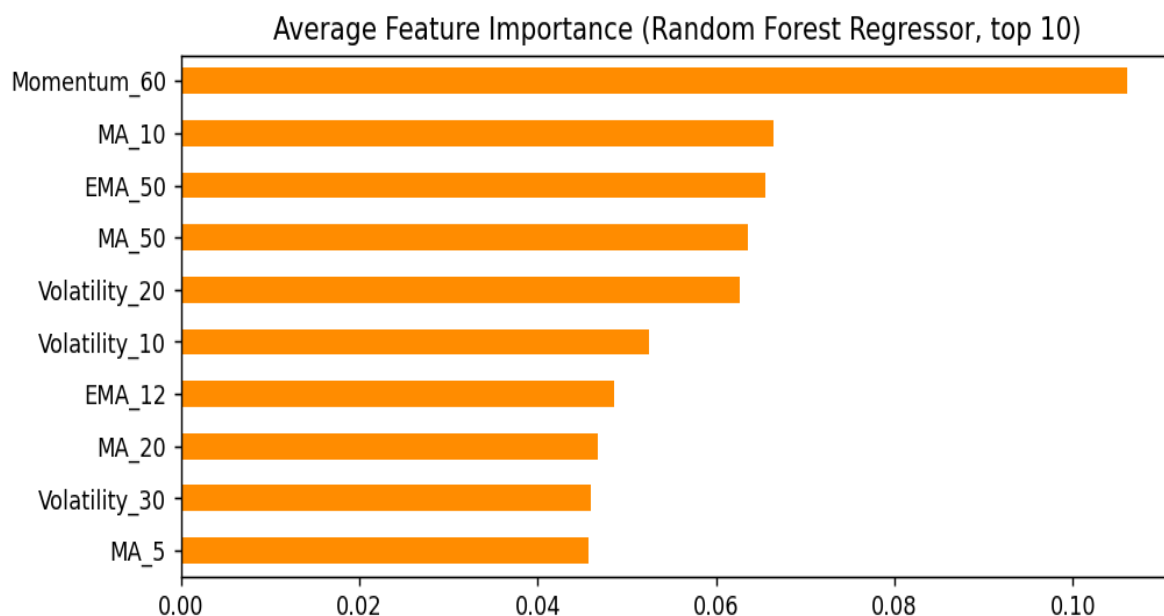


Figure 7.1 — Average feature importance across the 8-stock sample. Medium-term moving averages (MA-20), short-term momentum, and volatility measures dominate, consistent with mean-reversion / momentum effects documented in equity markets.

7.2 Portfolio Rationale

For every recommended portfolio, the dashboard exposes an **'Explainability'** panel stating: the optimization objective used, the eligible risk level(s) for the profile, the single-stock weight cap, and the per-stock historical return/volatility/Sharpe/risk-level statistics that justified each holding's inclusion (Section 6).

7.3 Risk Classification Transparency

Each stock's qualitative risk label (Low/Medium/High) is derived directly from a transparent, fixed volatility threshold (Section 5), and the dashboard displays the underlying annualized volatility number alongside the label so users can see exactly why a stock was classified the way it was.

7.4 Anomaly Explanations

Each flagged anomaly is reported with the underlying statistic that triggered it (e.g. "Volatility sharp drop (return -8.6%, z-score -4.1)" or "Unusual trading volume (z-score 3.8)"), giving users a direct, human-readable reason for the flag rather than an opaque score.

8. Key Insights

- Across the full NIFTY-50 universe (5-year lookback), the average annualized return is 15.1% with average annualized volatility 37.1% and average Sharpe ratio 0.21.
- Financial Services and Consumer Goods names generally offered the best risk-adjusted returns over the sample period, while PSU Energy names (IOC, ONGC, GAIL, Coal India) were structural underperformers with negative Sharpe ratios.
- Short-horizon (5-day) return prediction from price/volume-derived technical indicators alone yields directional accuracy only modestly above 50% and negative R^2 , reinforcing that technical signals provide, at best, a weak edge — consistent with semi-strong market efficiency. The platform reports this honestly rather than overstating predictive power.

- On RELIANCE, the anomaly detector flagged 151 unusual days and 4 major drawdown episode(s) over the full history, the largest being a 79% decline during the 2008–2012 period.
- The mean-variance optimizer produces intuitively sensible portfolios: the Conservative portfolio concentrates in a small number of low-volatility FMCG names (HUL, Nestlé India) with the lowest volatility and a respectable Sharpe ratio; the Balanced portfolio diversifies across 7–8 names spanning FMCG, financials and select cyclicals for the best risk-adjusted return; the Aggressive portfolio tilts toward higher-return, higher-volatility names (Bajaj Finserv, metals, Titan) for maximum expected growth.
- Overall, the platform demonstrates a complete pipeline from raw OHLCV data to actionable, explainable investment intelligence — while being explicit about the limitations of short-horizon price forecasting using only the permitted historical dataset.

Limitations & Future Work

- Models use only price/volume-derived features; incorporating fundamentals or sentiment (disallowed by the constraints of this challenge) could materially improve predictive power.
- The 5-year lookback for portfolio construction is a fixed design choice; rolling/walk-forward re-optimization could better reflect changing market regimes.
- Deep learning (LSTM/Transformer) sequence models were intentionally left out of this prototype in favor of interpretable classical ML, but represent a natural extension.